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2026 IEEE 5th World Conference on Applied Intelligence and Computing : Submission (3595) has been created.

1 message

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16 May 2026 at 18:23

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The following submission has been created.

Track Name: AIC2026

Paper ID: 3595

Paper Title: Towards Anatomy-Agnostic MRI Artifact Detection Using Unsupervised Learning

Abstract:

Artifacts in Magnetic Resonance Imaging (MRI) are distortions or anomalies that can arise from several factors including patient movement, hardware issues, physical tissue properties as well as acquisition related factors. These errors often render the scan useless and can sometimes be even confused with pathology. Recent advancements in medical imaging deep learning have led to significant progress in several medical imaging tasks, such as image classification, object detection, segmentation, registration, etc. Deep neural networks are known to be sensitive to image quality, leading to inaccurate results when fed with slices corrupted by artifacts. Existing artifact detection methods are built primarily on supervised methods, that require an enormous amount of well-annotated data. Such labelled datasets are time-consuming, subjective, scarce, and can render a lot of unlabeled data unutilized. Additionally, a lot of methods focus purely on a single sequence or anatomy, which can make models unable to explicitly disentangle anatomical features from artifacts. In accordance with these limitations, we propose a systematic evaluation of supervised, unsupervised, and self-supervised learning paradigms for MRI artifact detection under a unified experimental framework. We explore supervised baselines such as ResNet-18 and EfficientNet-B0, alongside unsupervised anomaly detection methods (e.g., autoencoders and PatchCore) and self-supervised representation learning approaches including SimCLR, DINO, and masked autoencoders. Our experiments are conducted on multi-cohort open MRI datasets to promote generalizability in acquisition settings and anatomy. Our findings suggest that learning anatomy-independent representations through label-free paradigms offers a scalable and practical pathway toward automated MRI quality control without requiring artifact-specific annotations.

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Research_MRIArtifact.docx (1 Mb, Sat, 16 May 2026 12:53:34 GMT)

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